Continuous-Time Markov Chains

From DTMCs to CTMCs

For DTMCs, transitions only made at discrete time steps. Want to generalize to transitions that can happen at any time, but keep the state space countable.

CTMC Definition

A Continuous-Time Markov Chain (CTMC) is a continuous-time stochastic process $\{X(t), t \geq 0\}$ such that $\forall s, t \geq 0$, and $\forall i, j, x(u)$,

$$P\{X(t+s) = j | X(s) = i, X(u) = x(u), 0 \leq u \leq s\}$$
$$= P\{X(t+s) = j | X(s) = i\}$$
$$= P\{X(t) = j | X(0) = i\} = P_{ij}(t)$$

The first equality is the Markov property, the second is stationarity.

Residence Times

Define τ_i to be the time until the CTMC leaves state i, given that the CTMC is currently in state i. Must have

$$P\{\tau_i > t + s | \tau_i > s\} = P\{\tau_i > t\}$$

But this means that τ_i is exponentially distributed.

Example

Consider a server system with a single backup. The time to failure is exponentially distributed with mean 86.4 hours and repair times are exponentially distributed with mean 2.5 hours. When a server is failed, a backup server is (instantaneously) started. This server is identical in failure/repair characteristics. A server can only fail when operating.

CTMC Model – Construction 1

- States: 0 (server operating, other server ready); 1 (server operating, other server failed); 2 (both servers failed)
- Time spent in state 0: $Exp(\frac{1}{86.4})$. The next state is 1 with probability 1.
- Time spent in state 1: $Exp(\frac{1}{86.4} + \frac{1}{2.5})$. The next state is 0 with probability $\frac{86.4}{88.9}$, otherwise it is state 2.
- Time spent in state 2: $Exp(\frac{1}{2.5} + \frac{1}{2.5})$. The next state is 1 with probability 1.
- So, time spent in state $i \sim Exp(\nu_i)$, upon exit go to state j with probability P_{ij}

CTMC Model – Construction 2

- Let's look at state 1
- Start two clocks, with timeouts given by $Exp(\frac{1}{86.4})$ and $Exp(\frac{1}{2.5})$. Whichever times out first makes the corresponding transition.
- Summarize this information in a *transition rate diagram* (on board).
- The two model constructions are equivalent – it should be intuitively obvious.

Limiting Probabilities

- Would like to solve for

$$\pi_j = \lim_{t \to \infty} P_{ij}(t),$$

the limiting probability of being in state j.

- One approach (in book) – convert to a DTMC problem and solve.
- Our approach is slightly different – we know that the number of transitions into state 1 differs from the number of transitions out of state 1 by at most one.

Limiting Probabilities

- If the number of transitions out of a state grows large as $t \to \infty$, then the rate of transitions into a state is equal to the rate of transitions out of a state.
- Thus,

$$\pi_1\left(\frac{1}{2.5}+\frac{1}{86.4}\right)=\frac{1}{86.4}\pi_0+\frac{2}{2.5}\pi_2$$

- There is nothing special about state 1, so we also get

$$\left(\frac{1}{86.4}\right)\pi_0=\left(\frac{1}{2.5}\right)\pi_1$$

$$\left(\frac{2}{2.5}\right)\pi_2=\left(\frac{1}{86.4}\right)\pi_1$$

Limiting Probabilities

As with DTMCs, no unique solution, but need $\pi_0 + \pi_1 + \pi_2 = 1$.

Global Balance Equations

The set of equations developed in this manner is known as the "global balance equations". The solution for this example is

$$\pi_0 = \frac{1}{1 + \frac{2.5}{86.4} + \frac{(2.5)^2}{2(86.4)^2}}$$

$$\pi_1 = \frac{2.5}{86.4}\pi_0$$

$$\pi_2 = \frac{(2.5)^2}{2(86.4)^2}\pi_0$$

Example

A system with fault detection is modelled as follows. When operating correctly, faults occur at a rate of 1 per hour, with fault times being exponentially distributed. With probability 0.99, a fault is detected and recovery takes an exponentially distributed period of time with mean 1 second. If the fault is not detected, recovery takes an exponentially distributed period of time with mean 5 minutes. Calculate the limiting probability that the system is operating correctly.

M/M/1 Queue

- single server
- arrivals follow a Poisson process, rate λ
- processing/service times follow $Exp(\mu)$ distribution (hence mean $1/\mu$)
- infinite waiting room/buffer

CTMC Model

- Let $X(t)$ be the number of jobs in the system at time t
- Global balance equations

$$\lambda\pi_0 = \mu\pi_1$$
$$(\lambda + \mu)\pi_n = \lambda\pi_{n-1} + \mu\pi_{n+1}, n \geq 1$$

Solving GBEs

$$\pi_1 = \frac{\lambda}{\mu}\pi_0$$

$$\pi_2 = \left(\frac{\lambda}{\mu} + 1\right)\pi_1 - \frac{\lambda}{\mu}\pi_0$$

$$= \frac{\lambda}{\mu}\pi_1 = \left(\frac{\lambda}{\mu}\right)^2 \pi_0$$

Solving GBEs

Similarly,

$$\pi_3 = \left(\frac{\lambda}{\mu} + 1\right)\pi_2 - \frac{\lambda}{\mu}\pi_1$$

$$= \left(\frac{\lambda}{\mu}\right)^3 \pi_0$$

Solving GBEs

In general,

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0$$

$$\pi_0 \left(\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu} \right)^n \right) = 1$$

$$\pi_0 = \frac{1}{\frac{1}{1-\rho}}$$

where $\rho = \frac{\lambda}{\mu} < 1$.

Finally,

$$\pi_n = (1 - \rho)\rho^n$$

The probability that the server is busy is $1 - \pi_0 = \rho$. For this reason, ρ is known as the *utilization*, or the *load*.

Some Numbers

Take $\rho = 0.1, 0.5, 0.9$.

	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.9}$
$\pi_0 + \pi_1 + \pi_2 + \pi_3$	.9999	.9375	.3439
$\pi_4 + \pi_5 + \pi_6 + \pi_7$	1×10^{-4}	.0586	.2256
$\pi_8 + \pi_9 + \pi_{10} + \pi_{11}$	1×10^{-8}	.00366	.1480
$\pi_{12} + \pi_{13} + \cdots$	1×10^{-12}	.000249	.2825

Main observation: things look very nonlinear in terms of ρ.

Expected number of jobs in system

$$
\begin{aligned}
E[N] &= \pi_0 \cdot 0 + \pi_1 \cdot 1 + \pi_2 \cdot 2 + \cdots \\
&= \rho(1-\rho) + 2\rho^2(1-\rho) + 3\rho^3(1-\rho) + \cdots \\
&= (1-\rho)\rho(1 + 2\rho + 3\rho^2 + \cdots) \\
&= (1-\rho)\rho\frac{d}{d\rho}(1 + \rho + \rho^2 + \rho^3 + \rho^4 + \cdots \\
&= (1-\rho)\rho\frac{d}{d\rho}\frac{1}{1-\rho} \\
&= (1-\rho)\rho\frac{1}{(1-\rho)^2} \\
&= \frac{\rho}{1-\rho}
\end{aligned}
$$

Mean Number in System

- Little's Law applies, so

$$E[T] = \frac{1}{\lambda}\frac{\rho}{1-\rho}$$
$$= \frac{1/\mu}{1-\rho} = \frac{1}{\mu-\lambda}$$

Example

Given an M/M/1 system, what is the minimum allowable processing rate such that an arrival rate of 2/9 jobs per minute, the mean response time is less than 9 minutes?

Example

Arrivals occur according to a Poisson process with rate 9 per minute. There are two servers, each with exponentially distributed processing times with mean 12 seconds. With probability p, an arriving job is assigned to server 1, otherwise it is assigned to server 2. What value of p minimizes the expected response time?

PASTA

- In the previous example, we implicitly used PASTA – Poisson Arrivals See Time Averages
- Proof in text, somewhat beyond the scope of the course
- Continue to use this property

Balance Equations Revisited

For the M/M/1 queue, what if we look at rates in both directions across a cut of the transition rate diagram?

Birth-Death Process

- Suppose that λ_n is the rate out of state n, to state $n+1$ and μ_n is the rate out of state n, to state $n-1$.
- This is known as a "Birth-Death Process" – the λ values are the birth rates and the μ values are the death rates.

Local Balance Equations

$$\lambda_0\pi_0 = \mu_1\pi_1$$
$$\lambda_1\pi_1 = \mu_2\pi_2$$
$$\vdots$$
$$\lambda_n\pi_n = \mu_{n+1}\pi_{n+1}$$

Local Balance Equations

Writing π_n in terms of π_0,

$$\pi_n = \frac{\lambda_0 \lambda_1 \lambda_2 \cdots \lambda_{n-1}}{\mu_1 \mu_2 \cdots \mu_n} \pi_0$$

So,

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \prod_{i=1}^{n} \frac{\lambda_{i-1}}{\mu_i}}$$

This is the general solution for Birth-Death processes.

M/M/1/N – finite buffer system

- arriving jobs (beyond N in system) “lost” or “blocked”
- lost jobs do not return at a later time
- transition rates:

$$\lambda_n = \lambda, \qquad n = 0,1,2,\dots,N-1$$
$$\mu_n = \mu, \qquad n = 1,2,3,\dots,N$$

M/M/1/N Solution

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0, \qquad 0 \le n \le N$$

$$\pi_0 = \frac{1}{\sum_{n=0}^{N} \left(\frac{\lambda}{\mu}\right)^n}$$

Rewriting

$$\pi_0 = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}}$$

Substitute π_0 in first expression to get π_n.

Blocking Probabilities

Blocking probability is the probability that the system is full, which is π_N here. The mean number of jobs lost per unit time is thus $\lambda\pi_N$. Here, for $N = 5$:

λ/μ	π_N
0.10	9×10^{-6}
0.50	0.016
0.75	0.072
1.00	0.166
2.00	0.508
5.00	0.800

Note for the $\lambda/\mu = 1.00$ entry, need to use L'Hopital's rule.

Blocking Probabilities

For $\lambda/\mu = 0.9$, vary buffer size:

N	π_N
1	0.474
2	0.299
3	0.212
5	0.126
10	0.051
20	0.014
100	2.7×10^{-6}

M/M/∞

Here, every job is immediately assigned its own processor (or each job its own server):

$$\pi_n = \left(\prod_{i=1}^{n} \frac{\lambda}{i\mu}\right)\pi_0$$

$$= \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n \pi_0$$

$$\pi_0 = \frac{1}{1 + \sum_{i=1}^{\infty} \frac{1}{i!}\left(\frac{\lambda}{\mu}\right)^i}$$

$$= e^{-\lambda/\mu}$$

M/M/∞

In the previous slide, the final equality comes from the fact that

$$\sum_{i=0}^{\infty} \frac{x^i}{i!} = e^x$$

M/M/∞

The final limiting distribution is thus

$$\pi_n = \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n e^{-\lambda/\mu}$$

- This is a Poisson distribution with mean $E[N]$.
- By Little's Law, $E[T] = 1/\mu$.
- Could get $E[T]$ by noting that there is no waiting in queue ($E\left[N_Q\right] = E\left[T_Q\right] = 0$).

M/M/c queue

System has c processors in parallel and a queue for additional jobs

$$\pi_n = \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n \pi_0, \qquad n \leq c$$

$$\pi_n = \frac{1}{c!\, c^{n-c}}\left(\frac{\lambda}{\mu}\right)^n \pi_0, \qquad n \geq c$$

$$\pi_0 = \left[1 + \sum_{n=1}^{c-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n + \frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \left(\frac{1}{1-\rho}\right)\right]^{-1}$$

where $\rho = \frac{\lambda}{c\mu} < 1$.

M/M/c queue

$$E[N_Q] = \left[\frac{\left(\frac{\lambda}{\mu}\right)^c \lambda\mu}{(c-1)!\,(c\mu-\lambda)^2}\right]\pi_0$$

$$E[T_Q] = \left[\frac{\left(\frac{\lambda}{\mu}\right)^c \mu}{(c-1)!\,(c\mu-\lambda)^2}\right]\pi_0$$

$$E[T] = \frac{1}{\mu} + E[T_Q]$$

$$E[N] = \lambda E[T]$$

Erlang-C formula

$$P\{queueing\} = \sum_{n=c}^{\infty} \pi_n$$

The sum evaluates to

$$\frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c \frac{1}{1-\rho}\pi_0$$

M/M/c/c

Loss system, no waiting. If a job arrives that finds all servers busy, lost. For $n \leq c$,

$$\pi_n = \left(\prod_{i=1}^{n} \frac{\lambda}{i\mu}\right)\pi_0$$

$$= \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n \pi_0$$

$$\pi_0 = \frac{1}{1 + \sum_{n=1}^{c} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n}$$

$$\pi_c = \frac{\frac{1}{c!}\left(\frac{\lambda}{\mu}\right)^c}{1 + \sum_{n=1}^{c} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n}$$

Client/Server Model

- Let the state be the number of jobs at the server
- Think time of clients exponentially distributed, rate λ, processing times exponentially distributed, rate μ
- No need to keep track of individual jobs, so does not matter what scheduling policy at CPU is

Client/Server Model

$$\pi_n = \left(\prod_{i=1}^{n} \frac{\lambda(M - i + 1)}{\mu} \right) \pi_0$$

$$\pi_n = \left(\left(\frac{\lambda}{\mu} \right)^n \frac{M!}{(M - n)!} \right) \pi_0$$

$$\pi_0 = \frac{1}{\sum_{n=0}^{M} \left(\frac{\lambda}{\mu} \right)^n \frac{M!}{(M - n)!}}$$

Example

Consider the client/server model with $\mu = 1$ and $\lambda = 0.1$. The throughput of the server is bounded above by μ (you should convince yourself of this). What value of M (number of users) leads to a throughput of 90 percent of this upper bound?

Example

An existing system consists of a single server. Requests arrive according to a Poisson process with rate 3 per hour. The processing times are exponentially distributed with mean 18 minutes. You are asked whether it would be beneficial to add a second (identical) processor. The two processors would share a common queue. The costs are as follows: operating costs of 25 dollars per hour per processor and cost of delay of 5 dollars per hour per request.

Replicating Requests – CTMC Model

Replication in Practice

- Google: send requests to multiple servers
- DNS queries
- Amazon EC2
- Distributed storage systems in general

Model

- Two-server model
- Arrival rate λ
- Processing rate μ
- Cancellation rate μ_c

Policies

- π_0: no replication (M/M/2 queue)
- π_∞: all requests replicated (need new CTMC model)

Example Dynamics: π_∞

- See r1.pdf

State Transition Diagram: π_∞

- See r2.pdf

Global Balance Equations: π_∞

$$\lambda\pi_0 = \mu_c\pi_{c,-1}$$
$$(\lambda + 2\mu)\pi_{s,0} = \lambda\pi_0 + \mu_c\pi_{c,0}$$
$$(\lambda + 2\mu)\pi_{s,j} = \lambda\pi_{s,j-1} + \mu_c\pi_{c,j} \qquad j = 1,2,\ldots$$
$$(\lambda + \mu_c)\pi_{c,-1} = 2\mu\pi_{s,0} + \mu\pi_{c,0}$$
$$(\lambda + \mu + \mu_c)\pi_{c,j} = 2\mu\pi_{s,j+1} + \lambda\pi_{c,j-1} + \mu\pi_{c,j+1} \qquad j = 0,1,2,\ldots$$

Example: $\lambda = 1, \mu = 1$

- Truncate replication system at 50 jobs
- M/M/2: E[T]=1.3333
- $\mu_c \to \infty$: $E[T] = 1$
- $\mu_c = 10$: $E[T] = 1.1095$
- $\mu_c = 2$: $E[T] = 1.6875$
- $\mu_c = 1$: $E[T] = 2.5714$

Example: $\lambda = 0.5, \mu = 1$

- Truncate replication system at 50 jobs
- M/M/2: E[T]=1.0667
- $\mu_c \to \infty$: $E[T] = 0.6667$
- $\mu_c = 10$: $E[T] = 0.6905$
- $\mu_c = 2$: $E[T] = 0.8036$
- $\mu_c = 1$: $E[T] = 0.9455$